

An Analysis of Student Opinion of Teaching Effectiveness Data

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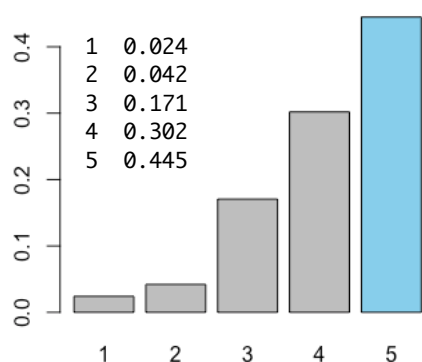
Math 258
Categorical Data Analysis
Fall 2019

Part I: Introduction

The main purpose of this report is to answer the question, “does the Student Evaluation of Teaching Effectiveness (SOTE) questionnaire accurately reflect the quality of a professor’s teaching effectiveness?” I will be analyzing data from the 2016 SOTE to see if there is a discernable relationship between the score a professor receives and other factors that are out of the professor’s control and have nothing to do with how good of a teacher they are.

The data consists of 86170 observation, each corresponding to the survey results of a particular student as well as information on that student and on the course for which the student is responding to the survey. The response variable of interest is “Q13” which is the student’s response to the question, “Overall, this instructor’s teaching was..” with their response being a number from 1 to 5, 1 being “very ineffective” and 5 being “very effective.”

Fig 1 - Proportion of Responses to Q13



The responses to Question 13 are highly negatively skewed. More than 44% of students gave their professors an overall effectiveness rating of “very effective,” which is a score of 5. For this reason, from the perspective of the evaluation committee, in order for a professor to be seen as “above average,” they need to receive lots of ratings of 5 on Q13 of the SOTE.

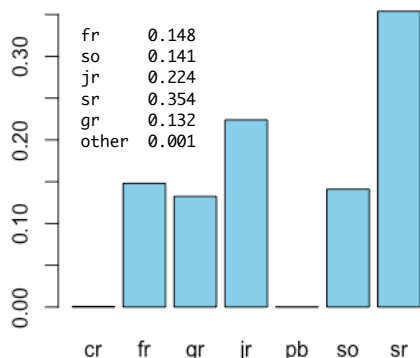
In addition to the response variable Q13, the data contains the following explanatory variables:

Variable	Type	Explanation
<i>semester</i>	Categorical	When was the class held? Either Fall 2016 or Spring 2017.
<i>grade</i>	Categorical	Official grade the student received in that class.
<i>grading</i>	Categorical	Grading scheme of that class, eg. credit/no credit ect.
<i>course.level</i>	Categorical	Lower division, upper division, etc.
<i>dept</i>	Categorical	Which department the class was offered in.
<i>college</i>	Categorical	Which college the class was offered in.
<i>student.level</i>	Categorical	Freshmen, sophomore, grad student, etc.
<i>enroll</i>	Continuous	Number of students enrolled in the class
<i>cap</i>	Continuous	Enrollment cap (maximum number of students that can enroll).
<i>registered</i>	Continuous	Number of students registered for the course
<i>Q14</i>	Categorical	Student reported estimate of what grade they expect to receive.
<i>Q15</i>	Categorical	Student reported grade level (freshmen, sophomore, etc.)

Table 1

Part II: Univariate Description of the Response Variables

Fig 2 - students' grade level



The proportion of students that are seniors is higher than any other grade, which is unexpected. We might expect to see a similar number of students in all grade levels. It is possible that some students finish their program in 5 or more years and both 4th year and 5th year students and beyond are marked as seniors. Or maybe seniors are more likely to answer the survey than other grade levels. The seniors and graduate students sometimes work as teaching assistant and may sympathize more with instructors given their teaching experience. Overall, they may give higher rating to instructors

Figure 4 shows the distribution of the number of students enrolled in each class. The data shows that class sizes vary between 1 and 460. We can see the distribution is skewed to the right, with most classes having about 20-50 students. When the class size is large, students get very little one on one time with the instructor. This may negatively impact their review of the instructor. On the other hand, when class size is small, students get a lot of one on one attention. It will be very surprising if an instructor with one student gets a bad review. In such a case, the instructor would know who gave the bad review. The larger the class size, the greater the feeling of anonymity the student has when answering the survey.

Fig 3 - distribution of class sizes

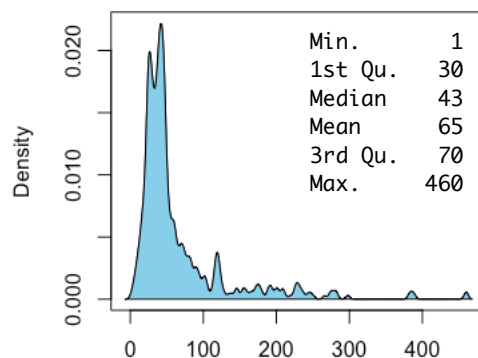
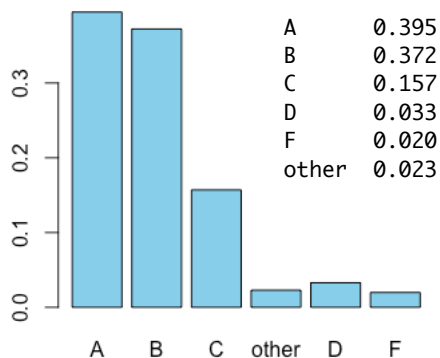


Fig 4 - students' reviewed grades



The majority of students who responded to the survey receive a grade of A or B. Maybe this is typical among all students, or maybe this is typical only among students who choose to respond to the survey. I suspect that students who receive a higher grade will be happier overall with their professor and thus give them a higher rating. This could account for the seemingly high number of A's and B's and also the high number of responses of 4 and 5 to Q13.

Part III: Marginal Associations

Table 2.

Expected Grade	Actual Grade				Total
	A	B	C	D or F	
A	0.728	0.254	0.017	0.001	1
B	0.259	0.614	0.123	0.004	1
C	0.074	0.420	0.472	0.034	1
D or F	0.046	0.178	0.512	0.263	1

Table 2 is the proportion of students who received each letter grade given what letter grade they predicted they would receive in Q14. If the student didn't respond, or if they chose or received something other than a letter grade such as credit/no credit, that student was omitted from the table.

A Chi-Squared test statistic of 50226 (critical value: 17), and a Goodman-Kruskal Gamma of 0.79 indicates there is a positive association between expected grade and actual grade. If one student expects a higher grade than another student, the first student is likely to *receive* a higher grade than the second student. This is not very surprising. Perhaps a more interesting question is *how good* are students at predicting what grade they will receive? The sample proportions highlighted on the diagonal of the table shows that the better the grade, the more likely a student will correctly predict it. Students are pretty good at predicting if they will get an A, yet most students who expect to get a D or F actually get a C.

A 95% Clopper-Pearson confidence interval for the probability that a student correctly predicts their letter grade is [.616, .623]. This is surprisingly low. Perhaps students aren't as good at predicting their actual grade as I might expect.

Table 3.

Expected Grade	Teacher Rating					Total
	5	4	3	2	1	
A	0.621	0.256	0.094	0.019	0.010	1
B	0.394	0.365	0.181	0.039	0.020	1
C	0.221	0.306	0.334	0.090	0.049	1
D or F	0.118	0.195	0.349	0.199	0.139	1

Table 3 is the proportion of ratings on a scale of 1-5 students gave the overall effectiveness of their professor given the letter grade they expected to receive. (1 = Very Ineffective; 2 = Ineffective; 3 = Somewhat Effective; 4 = Effective; 5 = Very Effective). Any response that wasn't a letter grade or wasn't a rating from 1-5 was removed.

A Chi-Squared test statistic of 12401 (critical value: 21), and a Goodman-Kruskal Gamma of 0.458 indicates there is a positive association between expected grade and teacher rating. The higher the grade a student expects to receive, the higher they would rate their teacher.

Table 4.

Actual Grade	Teacher Rating					Total
	5	4	3	2	1	
A	0.549	0.292	0.119	0.026	0.014	1
B	0.433	0.321	0.180	0.042	0.024	1
C	0.328	0.321	0.243	0.067	0.040	1
D or F	0.249	0.288	0.307	0.098	0.058	1

Table 4 shows the proportion of ratings students gave their professor given the letter grade they actually received. Any grade that wasn't a letter grade or any rating that wasn't a rating from 1-5 was removed.

A Chi-Squared test statistic of 4278 (critical value: 21), and a Goodman-Kruskal Gamma of 0.279 indicates there is a positive association between expected grade and teacher rating. However, it seems not as strong as the association between *expected* grade and teacher rating. It seems that the grade a student *expects* to receive has more effect on their rating than what grade they *actually* receive.

Fig 5 - proportion of ratings of 5

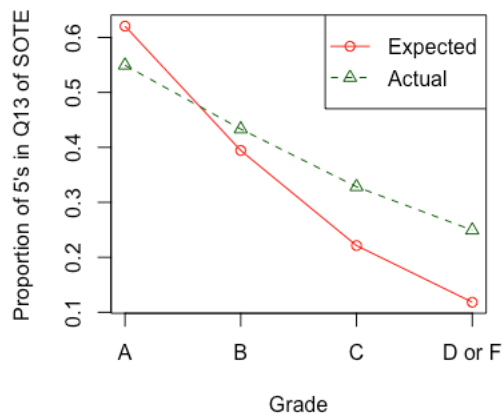


Figure 5 illustrates the difference between the effect of expected grade and actual grade on teacher rating. For both expected grade and actual grade, a higher grade is associated with a higher probability of giving a rating of 5 (“very effective”). However, we can see that the effect is much more pronounced for expected grade. It seems that when rating their teacher’s effectiveness, what grade the student *believes* they will get is more important than what grade they *actually* end up getting.

A chi-squared test statistic of 17.3 (critical value: 3.8) reveals there is an association between the semester the class took place in and whether the student gave a rating of 5 or not to the overall effectiveness of the professor. **Table 5** shows the standardized Pearson residuals from the chi-squared test, which gives some insight into the nature of the association. It appears that in the fall, fewer students gave a rating of 5 than we would expect to see if semester and rating were independent. Perhaps at the end of the spring semester students are in a better mood because they’re about to go on summer vacation, whereas at the end of the fall semester, students are stressed from the holiday season, cold weather, shorter amount of daylight, etc.

Table 5.

Semester	Rating	
	5	1-4
Fall	-4.2	4.2
Spring	4.2	-4.2

Table 6.

Student Level	Rating	
	5	1-4
Freshman	-14.0	14.0
Sophomore	-6.7	6.7
Junior	2.1	-2.1
Senior	9.9	-9.9
Graduate	5.1	-5.1

A chi-squared test statistic of 296.4 (critical value: 9.5) reveals there is an association between the student’s grade level and whether the student gave a rating of 5 or not to the overall effectiveness of the professor. **Table 6** shows the standardized Pearson residuals which shows which cells contributed most to the chi-squared test statistic being significant. It appears that freshmen are the harshest when rating their professors and seniors are the most generous. A 95% confidence interval for the difference of proportions between freshmen and seniors is [0.07, 0.09].

If we had a group of 100 seniors and 100 freshmen, we would expect about 7 to 9 more of the seniors to give a rating of 5 to Q13 of the SOTE than freshmen who give a rating of 5. This difference of 9 ratings to Q13 could very well be the difference between a professor getting a promotion or not.

The question I originally set out to investigate seems to have been answered. A student’s response to Q13 of the SOTE is most certainly associated with certain characteristics about the student and course which have nothing to do with the actual effectiveness of the professor. The question then remains: is there anything the professor can do to maximize the chance they will receive higher ratings? In the remaining section of this report I will construct a logistic regression model to see which combination of factors have the greatest effect on the response variable.

Part IV: Logistic Regression Model

As I briefly discussed earlier, from the perspective of the evaluation committee, a rating of 5 will be seen as “above average” and a rating of 4 or less will be seen as “below average.” I have first-hand experience with something like this in the real world. I used to sell cars at a Kia dealership. Every time we sold a new car, Kia would survey the customer. They would rate their satisfaction with our service on a scale from 1-5. Any score less than 5 was seen as a failure and resulted in scrutiny from management. We were even trained to tell our customer something like “You’ll be getting a call from Kia to rate your experience with me. Any score less than a 5 is seen as a failure on my part, so if you can’t give me a score of 5, please let me know what I can do to improve your experience.”

The evaluation committee is interested in separating professors into those who are deserving of a raise and those who are not. And perhaps it’s reasonable to draw the distinction that a score of 5 is deserving of a raise, but a score of 4 isn’t, especially considering how the data tends to be skewed towards scores of 5 (meaning that anything less than a 5 can be seen as a “failure”). For this reason, the response variable will be coded as binary; 1 if $Q13 = 5$, and 0 if $Q13 = 1, 2, 3$, or 4. The goal of my binary logistic regression model is to make suggestions to a professor facing a possible raise as to how they might maximize the probability of each student they teach giving them a rating of 5 on Q13 of the SOTE.

Before fitting a generalized linear model (GLM), I wanted to identify potential sources of multicollinearity. I suspect many predictors are highly correlated with one another, so they would be redundant in my model. For example, earlier I discussed expected grade and actual grade. I suspect these two variables are highly correlated, and I should only use one of them in my model. As I showed earlier, expected grade is a much better predictor of the response variable $Q13$ than actual grade. I confirmed this by fitting two GLMs, one with expected grade as the only predictor and one with actual grade as the only predictor, and compared the AIC values. The model with expected grade had a lower AIC (113603.2 vs 119808.1). Furthermore, it wouldn’t even really make sense to predict professor rating with actual grade, because the students fill out the SOTE before they receive their actual grades. Thus, I chose to include expected grade ($Q14$) in my model, but not actual grade.

Next I took a look at the variables *cap*, *enroll*, and *registered*. These three variables are all very highly correlated as we see in the correlation matrix in **Table 7**. I fit three GLMs, one for each predictor and compared the AIC values. The model with *cap* as the predictor had the lowest AIC, so I decided to include *cap* in the model and omit the other two variables. This works out well logically, as well, because *cap* is something that the professor has more control over than enrollment or registration. So including *cap* in the model may lead to more actionable recommendations.

Table 7.

	<i>cap</i>	<i>enroll</i>	<i>registered</i>
<i>cap</i>	1	0.9613	0.9603
<i>enroll</i>	0.9613	1	0.9998
<i>registered</i>	0.9603	0.9998	1

Table 8.

	<i>course.level</i>	<i>student.level</i>	<i>Q15</i>
<i>course.level</i>	1	0.92	0.88
<i>student.level</i>	0.92	1	0.93
<i>Q15</i>	0.88	0.93	1

The next set of variables I examined was student's official level, students self-reported level (*Q15*), and course level. **Table 8** is a matrix of Goodman-Kruskall Gammas between the three variables, which shows that they are all very highly

associated with one another, so I should only use one in my model. After fitting three models, one for each predictor, the one with the lowest AIC was course level. I decided to include *course.level* in my model and exclude the other two variables.

Finally, I examined *dept* and *college*. Each department belongs to one and only one college, thus the variable *college* contains redundant information. I decided to include *dept* but not *college*, because the variable *dept* contains more information. However, there were simply too many levels of *dept*, so I decided to combine some departments together to help to simplify things a little bit. I ended up going from 54 departments to 18 departments (see appendix on page 14 for a full list of departments I combined into one).

After Identifying which variables would likely be in my final model, I started by fitting several models with only two predictors in order to get a better idea how the predictors affect the probability of receiving a 5 on *Q13*, and to potentially identify the need for interaction terms.

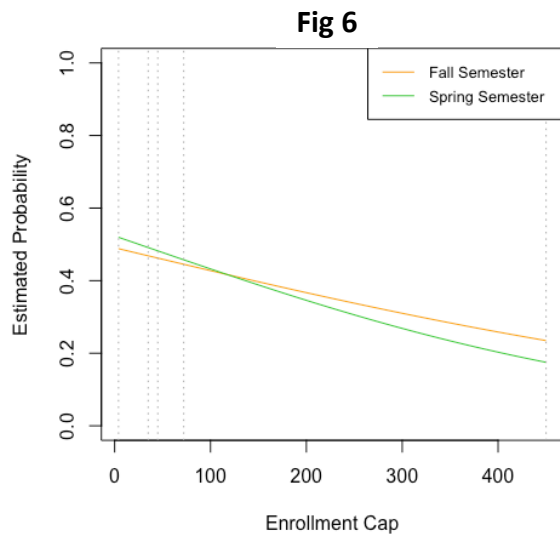


Table 9.

	Enrollment Cap		
Semester	35	45	72
Fall	0.468	0.462	0.445
Spring	0.491	0.482	0.458

Figure 6 and **Table 9** show that to maximize the probability of receiving a rating of 5, it may be wise for a professor to teach as small of a class as possible and preferably in the spring semester. The lines intersect, indicating interaction between the predictors. If you must teach a very large class (>116), it's better to do so in the fall semester, even though overall, it's better to teach in the spring semester. The dotted vertical lines designate the 1st quartile, median, and 3rd quartile of the distribution of the variable *cap*.

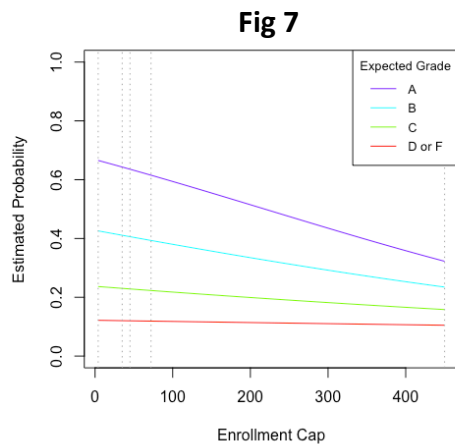


Table 10.

Expected Grade	Enrollment Cap		
	35	45	72
A	0.643	0.636	0.615
B	0.411	0.406	0.393
C	0.230	0.228	0.223
D or F	0.120	0.120	0.119

Figure 7 and **Table 10** show that to maximize the chance of receiving a rating of 5, professors should make as many students as possible feel like they're going to receive an A. Perhaps they should constantly remind students that they grade on a curve, offer exam corrections, etc. The grade the students *believe* they will receive is what matters. The lines do not intersect, implying that regardless of class size, it's better for your students to believe they will receive a higher grade (i.e. no interaction). We also see that the higher the grade they expect to receive, the larger the effect the class size has. For students who expect to receive a D or an F, class size basically doesn't matter at all. The probability they give a rating of 5 remains the same.

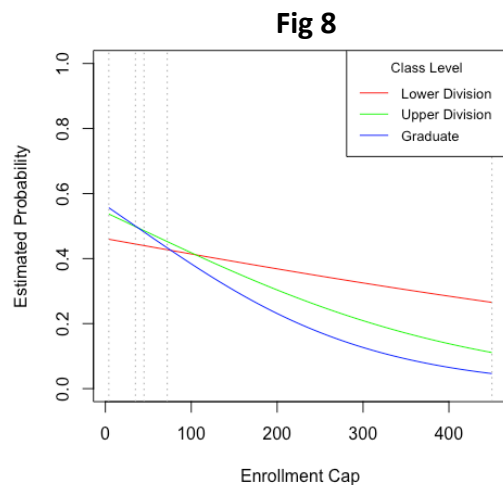


Table 11.

Class Level	Enrollment Cap		
	35	45	72
Lower Division	0.445	0.440	0.427
Upper Division	0.498	0.486	0.452
Graduate	0.500	0.482	0.433

Figure 8 and **Table 11** show that for normal sized classes, teaching graduate level or upper division classes will typically maximize your chance of getting a rating of 5 on Q15. It seems that for very large class sizes, it may be better to teach lower division classes, however I am doubtful about the accuracy of these estimates because classes of this size (larger than 100) are almost always lower division classes. I can't imagine a graduate level class exists with an enrollment cap above 100.

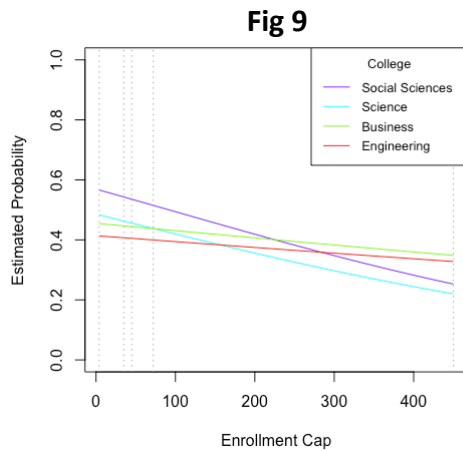


Table 12.

Enrollment Cap			
College	35	45	72
Social Sciences	0.543	0.535	0.515
Science	0.463	0.456	0.438
Business	0.447	0.444	0.438
Engineering	0.407	0.405	0.400

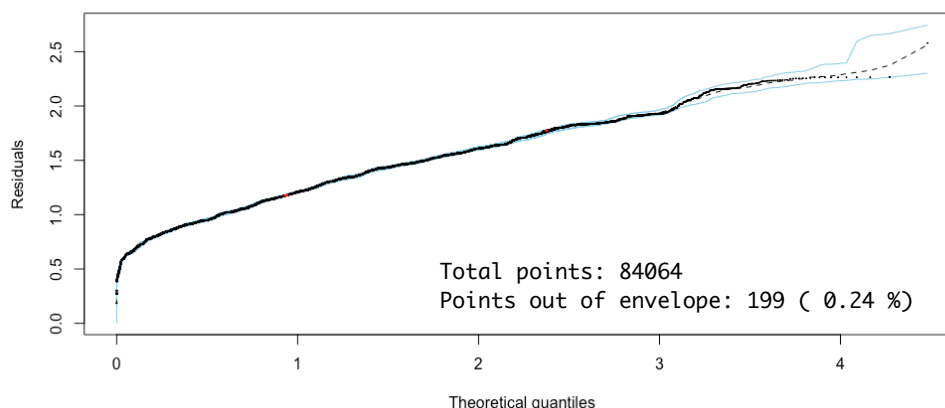
For **Figure 9** and **Table 12**, I decided to use the predictor *college* instead of *dept* for simplicity's sake. Even after condensing the departments down to 18, if I plotted all 18, it would be a mess. We observe that for normal sized classes, social sciences professors are rated the most generously and engineering professors are rated the most harshly. For very large classes, business professors are rated the most generously and science professors are rated the most harshly. Class size has the highest impact on the probability of receiving a 5 for social sciences professors, and it has the lowest effect on engineering professors.

Next I fit my full model, omitting those variables I identified as redundant on pages 5-6, and including interaction terms I identified as meaningful on pages 6-7. For a full R output of coefficients and significance tests, please see page 16-17 of the appendix. After fitting my full model, I performed model diagnostics to see if our model assumptions are reasonable. Such assumptions include:

1. Our response variable follows a binomial distribution.
2. Our predictors have a linear relationship with the log-odds of the response variable.
3. Our observations are independent of one another.
4. Our residuals are uncorrelated.

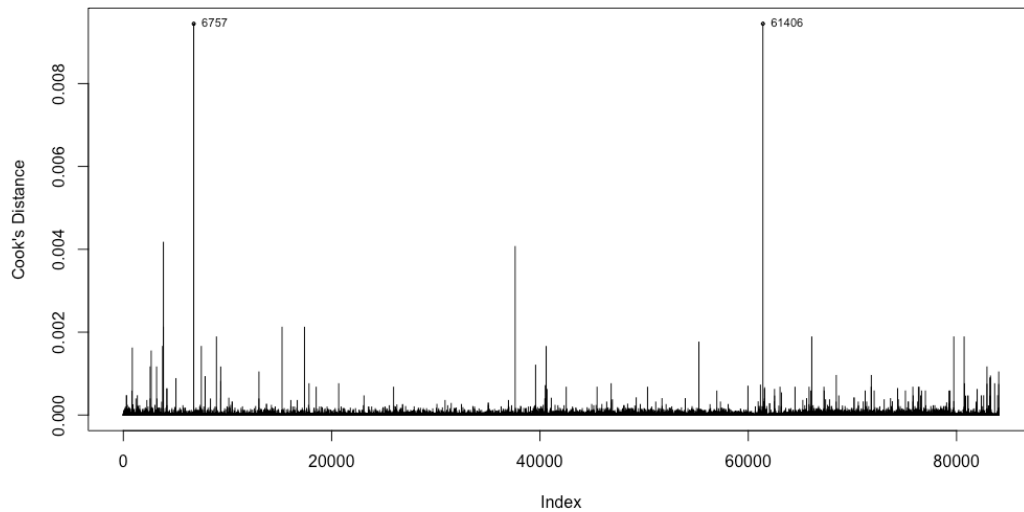
Part V: Model Diagnostics

Fig 10 – half-normal plot



From **Figure 10**, we see that only 199 out of 84064 points are outside of the envelope, meaning that, for the most part, the quantiles of our residuals fall within the boundaries of the simulations. This shows that our binomial logistic regression model is reasonable for this data.

Fig 11 – Cook’s Distance



In the Cook’s Distance plot, we see two observations stand out above the rest. Though it’s unclear whether these are actually influential points or not. Some authors suggest a rule of thumb of 1 as a cut off for influential observations using Cook’s Distance. 0.008 is well below 1. However, other authors have suggested using $4/n$ as a rule of thumb where n is the number of observations, in which case, since n is so large, nearly half of the observations would be considered influential!

Interestingly, the two observations with the highest Cook’s D came from the same exact class. They were both from a graduate level meteorology class in fall 2016, with 7 students and an enrollment cap of 10. Probably the most unusual thing about these observations is the grading was CMC, which means mandatory credit/no credit. Only 7 observations included CMC grading, which is only about 0.008% of all observations. This may account for the potentially influential nature of these two observations.

Table 13.

Observed	Predicted		Total
	Rating of 5	Rating of 1-4	
Rating of 5	22850	15077	37927
Rating of 1-4	14219	31918	46137

Sensitivity = 0.60
Specificity = 0.69

Fig 12 - ROC curve

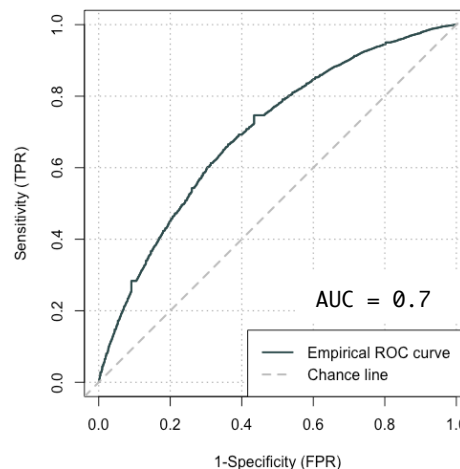


Table 13 Shows that when the cutoff is set to 0.5, our model correctly predicts a rating of 5 on *Q13* about 60% of the time, and it correctly predicts a rating which is *not* 5 about 69% of the time. The predictive power of the model is not outstanding, but it’s not completely terrible either. **Figure 12** shows an AUC of 0.7, which is similarly not great, but not terrible.

Part VI: Conclusion

My model seems to have reasonable predictive power and seems to satisfy my model assumptions reasonably well. I believe this model is adequate. In conclusion, for professors seeking a promotion, there are several things they should keep in mind when deciding which classes to teach. They need to aim for a score of 5 on Q13 of the SOTE. Otherwise, their performance will be seen as “below average” by the evaluation committee, and thus not deserving of a raise. If possible, the professor should teach a very, very small upper division class in the social sciences and in the fall semester. They should also make an effort to make the students believe they will receive a high grade, whether or not that’s true.

If possible, the professor should try to teach to education students. Not surprisingly, when a class is meant to be taken by education majors, the students rate their professors very generously. For a student who expects to receive an A in an upper division education class with a cap of 10, in the fall semester, my model predicts that they will rate their professor a 5 about 83% of the time. (95% C.I. = [0.811, 0.848]). That’s *much* better than an overall probability of about 44.5%. (95% C.I. = [0.441, 0.448]). If that very same student was taking an upper division math or stats class instead of an education class, my model predicts that student will rate their professor a 5 about 70% of the time. (95% C.I. = [0.689, 0.708]). That’s still pretty good.

Math students are not the most generous when it comes to rating their professors. Besides education, the following departments are preferable to math in terms of how generous the students are with their ratings: sociology, anthropology, history, and psychology. I find it interesting that these generally fall under the umbrella of social sciences.

Students in the following departments are less generous than math and stats students when rating their professors: technology, astronomy, biology, business, and economics. Students in the following departments show no difference to, or rate their professors very similarly to math and stats students: natural sciences, chemistry, communication, geography, meteorology, physics, and urban planning.

Of course, none of this is to say that the quality of one’s teaching doesn’t matter at all. Of course it does. But beyond a professor’s efforts and abilities, there are certain qualities about a particular student or course which will affect the professor’s SOTE scores. My hope with this report was to highlight some of the most important of these factors to empower any professor who reads this with his or her best chance of receiving a promotion.

Part VII: References

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- Schreiber-Gregory & Jackson, Logistic and Linear Regression Assumptions: Violation Recognition and Control,
https://www.lexjansen.com/wuss/2018/130_Final_Paper_PDF.pdf

Part VIII: Appendix

```
library(car)
library(readxl)
library(RCfit)
library(hnp)

###introduction/univariate analysis###
sote=read_xlsx("sote-data-part.xlsx")
View(sote)
str(sote)
#variable 1: semester
sote$semester=as.factor(sote$semester)
barplot(table(sote$semester)/nrow(sote))
#variable 2: grade
sote$grade=as.factor(sote$grade)
levels(sote$grade)=c("A", "A", "A", "B", "B", "B", "C", "C", "C", "other", "D", "D", "D", "F", "other", "other",
"other", "other", "other")
barplot(table(sote$grade)/nrow(sote),main="Fig 4 - students' reviewed grades",col="skyblue")
round(table(sote$grade)/nrow(sote),3)
#variable 3: grading
sote$grading=as.factor(sote$grading)
barplot(table(sote$grading)/nrow(sote))
#variable 4: course.level
sote$course.level=as.factor(sote$course.level)
barplot(table(sote$course.level)/nrow(sote))
#variable 5: dept
sote$dept=as.factor(sote$dept)
barplot(table(sote$dept)/nrow(sote))
#variable 6: college
sote$college=as.factor(sote$college)
barplot(table(sote$college)/nrow(sote))
#variable 7: student.level
sote$student.level=as.factor(sote$student.level)
barplot(table(sote$student.level)/nrow(sote),main="Fig 2 - students' grade level",col="skyblue")
round(table(sote$student.level)/nrow(sote),3)
#variable 8: enroll
plot(density(sote$enroll),main="Fig 3 - distribution of class sizes")
polygon(density(sote$enroll),col="skyblue")
summary(sote$enroll)
length(subset(sote$enroll,sote$enroll>20&sote$enroll<50))/length(sote$enroll)
#variable 9: cap
plot(density(sote$cap))
#variable 10: registered
plot(density(sote$registered))
#variable 11: Q13
sote$Q13=as.factor(sote$Q13)
round(table(sote$Q13)[1:5]/nrow(sote),3)
barplot(table(sote$Q13)[1:5]/nrow(sote),main="Fig 1 - Proportion of Responses to
Q13",col=c("grey", "grey", "grey", "grey", "skyblue"))
#variable 12: Q14
sote$Q14=as.factor(sote$Q14)
barplot(table(sote$Q14)/nrow(sote))
#variable 13: Q15
sote$Q15=as.factor(sote$Q15)
barplot(table(sote$Q15)/nrow(sote))
```

```

###marginal associations###
##exploring expected grade and actual grade##
levels(sote$grade)=c("A","B","C","other","DorF","DorF")
levels(sote$Q14)=c("A","B","C","DorF","other",NA)
exp.vs.grade=table(sote$grade,sote$Q14)[-4,-5]
round(exp.vs.grade/rowSums(exp.vs.grade),3)
chisq.indep(exp.vs.grade)
gk.gamma(exp.vs.grade)
#clopper pearson ci
s=sum(diag(exp.vs.grade))
n=sum(exp.vs.grade)
qbeta(1-.05/2,s,n-s+1,lower.tail = F)
qbeta(.05/2,s+1,n-s,lower.tail = F)
exp.vs.rating=table(sote$Q14,sote$Q13)[-5,5:1]
chisq.indep(exp.vs.rating)
gk.gamma(exp.vs.rating)
round(exp.vs.rating/rowSums(exp.vs.rating),3)
grade.vs.rating=table(sote$grade,sote$Q13)[-4,5:1]
chisq.indep(grade.vs.rating)
gk.gamma(grade.vs.rating)
round(grade.vs.rating/rowSums(grade.vs.rating),3)
plot(exp.vs.rating[,1]/rowSums(exp.vs.rating),xaxt="n",xlab="Grade",ylab="Proportion of 5's in
Q13 of SOTE",col="red",main="Fig 5 - proportion of ratings of 5")
axis(1,at=1:4,labels = c("A","B","C","D or F"))
points(grade.vs.rating[,1]/rowSums(grade.vs.rating),col="darkgreen",pch=2)
lines(exp.vs.rating[,1]/rowSums(exp.vs.rating),col="red")
lines(grade.vs.rating[,1]/rowSums(grade.vs.rating),col="darkgreen",lty=2)
legend("topright",legend=c("Expected","Actual"),col=c("red","darkgreen"),pch=1:2,lty=1:2)
round(chisq.test(exp.vs.rating)$stdres,1)
##exploring semester##
levels(sote$semester)=c("fall","spring")
sem=table(sote$semester,sote$Q13)[,5:1]
sem[1,2]=sum(sem[1,2:5])
sem[2,2]=sum(sem[2,2:5])
sem=sem[,1:2]
colnames(sem)=c("5","1:4")
chisq.indep(sem)
round(chisq.test(sem)$stdres,1)
##exploring student level##
level=sote$student.level
levels(level)=c(NA,"fr","gr","jr",NA,"so","sr")
rating=sote$Q13
levels(rating)=c("1:4","1:4","1:4","1:4","5",NA,NA)
level=table(level,rating)[c(1,4,3,5,2),]
round(level/rowSums(level),3)
chisq.indep(level)
round(chisq.test(level)$stdres,1)
m=level[c(1,4),][2:1,2:1]
t1hat=m[1,1]/sum(m[1,])
t2hat=m[2,1]/sum(m[2,])
DPhat=t1hat-t2hat
var=t1hat*(1-t1hat)/sum(m[1,])+t2hat*(1-t2hat)/sum(m[2,])
DPhat-qnorm(.975)*sqrt(var)
DPhat+qnorm(.975)*sqrt(var)

```

```

####model building###
##making Q13 binary
levels(sote$Q13)=c(F,F,F,F,T,NA,NA)
sote$Q13=as.logical(sote$Q13)
##variable selection##
levels(sote$course.level)=c("GR","LD",NA,"UD")
levels(sote$Q15)=c("1","2","3","4","5","6","7",NA)
sotemodel1=glm(Q13~.,data=sote,family = "binomial")
summary(sotemodel1)
vif(sotemodel1)
alias(sotemodel1)
sotemodel2=glm(Q13~semester+grade+grading+course.level+dept+student.level+enroll+cap+registered+Q
14+Q15,data=sote,family = "binomial")
summary(sotemodel2)
vif(sotemodel2)
summary(glm(Q13~grade,data=sote))$aic
summary(glm(Q13~Q14,data=sote))$aic #winner
round(cor(data.frame(cap=sote$cap,enroll=sote$enroll,registered=sote$registered)),4)
summary(glm(Q13~enroll,data=sote))$aic
summary(glm(Q13~registered,data=sote))$aic
summary(glm(Q13~cap,data=sote))$aic #winner
l1=sote$course.level
l1=ordered(l1,levels=c("LD","UD","GR"))
l2=sote$student.level
levels(l2)=c(NA,"fr","gr","jr",NA,"so","sr")
l2=ordered(l2,levels=c("fr","so","jr","sr","gr"))
l3=sote$Q15
levels(l3)=c("1","2","3","4","5",NA,NA)
l3=ordered(l3)
round(gk.gamma(table(l1,l2)),2)
round(gk.gamma(table(l1,l3)),2)
round(gk.gamma(table(l2,l3)),2)
summary(glm(Q13~student.level,data=sote))$aic
summary(glm(Q13~Q15,data=sote))$aic
summary(glm(Q13~course.level,data=sote))$aic #winner
levels(sote$dept)
dept=as.character(read.table("dept.txt",header=F)[,1])
levels(sote$dept)=dept
#
# new dept:      old depts:
# anthropology   ANTH
# astronomy      ASTR
# biology         BIOL, BME, ENT, MICR
# business       BUS1, BUS2, BUS3, BUS4, BUS5, PADM
# chemistry      CHE, CHEM
# communication  COMM
# economics      ECON
# edu            MTED, SCED, SCI, SOCS, SSER
# geography      GEOG
# history        HIST
# math           MATH, STAT
# meteorology    METR
# nature         BOT, ENVS, GEOL, ZOOL
# physics        PHYS
# psychology     PSYC
# sociology      AAS, AFAM, GLST, HUMC, MAS, ORGS, POLS, SOCI, SSCI, WOMS
# technology     AE, AVIA, CE, CMPE, CS, EE, ENGR, ISE, MATE, ME, TECH
# urban.planning URBP
#

```

```

##model with cap and semester
sote1=glm(Q13~semester*cap,data=sote,family = "binomial")
summary(sote1)
x1 <- quantile(sote$cap, c(0.25, 0.5, 0.75))
x2 <- rep(levels(sote$semester), each = 3)
probs <- round(matrix(predict(sote1,newdata = data.frame(cap = x1, semester = x2, row.names =
NULL),type = "response"), nrow = 2, byrow = TRUE,dimnames = list(c("Fall","Spring"),c("Q1",
"Q2","Q3"))),3)
x=seq(min(sote$cap),max(sote$cap),by=.1)
y=c(0,1)
plot(0,0,col = "white", lwd = 1,ylim=y,xlim=c(min(sote$cap),max(sote$cap)),xlab="Enrollment
Cap",ylab="Estimated Probability",main="Fig 6")
curve(predict(sote1, newdata = data.frame(cap = x, semester = "fall", row.names = NULL),type =
"response"),ylim = y,col = "orange", lwd = 1,add=T)
curve(predict(sote1, newdata = data.frame(cap = x, semester = "spring", row.names = NULL),type =
"response"),ylim = y,col = "limegreen", lwd = 1,add=T)
abline(v=summary(sote$cap)[-4],lty="dotted",col="darkgrey")
legend("topright",legend=c("Fall Semester","Spring
Semester"),col=c("orange","limegreen"),lty=1,cex=.8)
predict(sote1, newdata = data.frame(cap = 115:116, semester = "fall", row.names =
NULL),type="response")
predict(sote1, newdata = data.frame(cap = 115:116, semester = "spring", row.names =
NULL),type="response")
##model with cap and grade
sote2=glm(Q13~Q14*cap,data=sote,family = "binomial")
summary(sote2)
x1 <- quantile(sote$cap, c(0.25, 0.5, 0.75))
x2 <- rep(levels(sote$Q14)[-5], each = 3)
probs <- round(matrix(predict(sote2,newdata = data.frame(cap = x1, Q14 = x2, row.names =
NULL),type = "response"), nrow = 4, byrow = TRUE,dimnames = list(c("A","B","C","D or F"),c("Q1",
"Q2", "Q3"))),3)
x=seq(min(sote$cap),max(sote$cap),by=.1)
y=c(0,1)
plot(0,0,col = "white", lwd = 1,ylim=y,xlim=c(min(sote$cap),max(sote$cap)),xlab="Enrollment
Cap",ylab="Estimated Probability",main="Fig 7")
curve(predict(sote2, newdata = data.frame(cap = x, Q14 = "A", row.names = NULL),type =
"response"),ylim = y,col = rainbow(4)[4], lwd = 1,add=T)
curve(predict(sote2, newdata = data.frame(cap = x, Q14 = "B", row.names = NULL),type =
"response"),ylim = y,col = rainbow(4)[3], lwd = 1,add=T)
curve(predict(sote2, newdata = data.frame(cap = x, Q14 = "C", row.names = NULL),type =
"response"),ylim = y,col = rainbow(4)[2], lwd = 1,add=T)
curve(predict(sote2, newdata = data.frame(cap = x, Q14 = "D or F", row.names = NULL),type =
"response"),ylim = y,col = rainbow(4)[1], lwd = 1,add=T)
abline(v=summary(sote$cap)[-4],lty="dotted",col="darkgrey")
legend("topright",title="Expected Grade",legend=c("A","B","C","D or
F"),col=rev(rainbow(4)),lty=1,cex=.8)
##model with cap and course level
sote3=glm(Q13~course.level*cap,data=sote,family = "binomial")
summary(sote3)
x1 <- quantile(sote$cap, c(0.25, 0.5, 0.75))
x2 <- rep(levels(sote$course.level), each = 3)
probs <- round(matrix(predict(sote3,newdata = data.frame(cap = x1, course.level = x2, row.names =
NULL),type = "response"), nrow = 3, byrow = TRUE,dimnames = list(c("GR","LD","UD"),c("Q1",
"Q2","Q3"))),3)
x=seq(min(sote$cap),max(sote$cap),by=.1)
y=c(0,1)
plot(pred.prob(x, a = alpha[1], b = beta),col = "white", lwd =
1,ylim=y,xlim=c(min(sote$cap),max(sote$cap)),xlab="Enrollment Cap",ylab="Estimated
Probability",main="Fig 8")

```

```

curve(predict(sote3, newdata = data.frame(cap = x, course.level = "LD", row.names = NULL),type =
"response"),ylim = y,col = rainbow(3)[1], lwd = 1,add=T)
curve(predict(sote3, newdata = data.frame(cap = x, course.level = "UD", row.names = NULL),type =
"response"),ylim = y,col = rainbow(3)[2], lwd = 1,add=T)
curve(predict(sote3, newdata = data.frame(cap = x, course.level = "GR", row.names = NULL),type =
"response"),ylim = y,col = rainbow(3)[3], lwd = 1,add=T)
abline(v=summary(sote$cap)[-4],lty="dotted",col="darkgrey")
legend("topright",title="Class Level",legend=c("Lower Division","Upper
Division","Graduate"),col=rainbow(3),lty=1,cex=.8)
##model with cap and college
sote4=glm(Q13~college*cap,data=sote,family = "binomial")
summary(sote4)
x1 <- quantile(sote$cap, c(0.25, 0.5, 0.75))
x2 <- rep(levels(sote$college), each = 3)
probs <- round(matrix(predict(sote4,newdata = data.frame(cap = x1, college = x2, row.names =
NULL), type = "response"), nrow = 4, byrow = TRUE, dimnames =list(c("bus","eng","sci","soc"),
c("Q1","Q2","Q3"))),3)
x=seq(min(sote$cap),max(sote$cap),by=.1)
y=c(0,1)
plot(0,0,col = "white", lwd = 1,ylim=y,xlim=c(min(sote$cap),max(sote$cap)),xlab="Enrollment
Cap",ylab="Estimated Probability",main="Fig 9")
curve(predict(sote4, newdata = data.frame(cap = x, college = "soc", row.names = NULL),type =
"response"),ylim = y,col = rainbow(4)[4], lwd = 1,add=T)
curve(predict(sote4, newdata = data.frame(cap = x, college = "sci", row.names = NULL),type =
"response"),ylim = y,col = rainbow(4)[3], lwd = 1,add=T)
curve(predict(sote4, newdata = data.frame(cap = x, college = "bus", row.names = NULL),type =
"response"),ylim = y,col = rainbow(4)[2], lwd = 1,add=T)
curve(predict(sote4, newdata = data.frame(cap = x, college = "eng", row.names = NULL),type =
"response"),ylim = y,col = rainbow(4)[1], lwd = 1,add=T)
abline(v=summary(sote$cap)[-4],lty="dotted",col="darkgrey")
legend("topright",title="College",legend=c("Social
Sciences","Science","Business","Engineering"),col=rev(rainbow(4)),lty=1,cex=.8)
##Full model
sote$grading=relevel(sote$grading,"GRD")
sote$dept=relevel(sote$dept,"math")
sotemodel3=glm(Q13~semester+grading+course.level+dept+cap+Q14+cap:semester+cap:course.level+cap:d
ept,data=sote,family = "binomial")
summary(sotemodel3)
#
# Call:
# glm(formula = Q13 ~ semester + grading + course.level + dept +
#       cap + Q14 + cap:semester + cap:course.level + cap:dept, family = "binomial",
#       data = sote)
#
# Deviance Residuals:
#   Min       1Q   Median       3Q      Max
# -1.8962 -1.0202 -0.6405  1.0878  2.5781
#
# Coefficients:
#               Estimate Std. Error z value Pr(>|z|)
# (Intercept)      7.157e-01  6.826e-02  10.484 < 2e-16 ***
# semesterspring    7.016e-02  2.293e-02   3.059 0.002217 **
# gradingANC        2.157e-01  9.903e-02   2.178 0.029397 *
# gradingCMC       -1.149e+00  8.637e-01  -1.330 0.183367
# gradingCNC       -2.895e-02  7.922e-02  -0.365 0.714781
# gradingGRN        4.418e-01  3.685e-01   1.199 0.230552
# course.levelLD    -2.368e-01  5.714e-02  -4.144 3.41e-05 ***
# course.levelUD     1.492e-01  5.501e-02   2.711 0.006702 **
# deptsociology     1.957e-01  7.002e-02   2.795 0.005185 **
# depttechnology    -4.354e-01  5.084e-02  -8.563 < 2e-16 ***

```

```

# deptanthropology      3.679e-01  1.595e-01  2.307 0.021055 *
# deptastronomy         -2.906e+00  1.295e+00 -2.243 0.024893 *
# deptbiology           -1.692e-01  6.434e-02 -2.630 0.008545 **
# deptnature            -4.057e-02  8.950e-02 -0.453 0.650346
# deptbusiness          -1.741e-01  6.044e-02 -2.881 0.003963 **
# deptchemistry         1.239e-01  7.774e-02  1.594 0.110900
# deptcommunication     -4.885e-01  3.231e-01 -1.512 0.130603
# depteconomics        -3.898e-01  1.044e-01 -3.735 0.000188 ***
# deptgeography         3.659e-02  4.321e-01  0.085 0.932509
# depthistory           6.554e-01  1.002e-01  6.545 5.96e-11 ***
# deptmeteorology      -3.832e-01  2.321e-01 -1.651 0.098798 .
# deptedu               8.230e-01  1.439e-01  5.721 1.06e-08 ***
# deptphysics           1.613e-01  8.453e-02  1.908 0.056432 .
# deptpsychology        2.190e-01  6.018e-02  3.638 0.000274 ***
# depturban.planning    2.929e-02  5.795e-01  0.051 0.959686
# cap                   -3.512e-03  1.007e-03 -3.489 0.000485 ***
# Q14B                  -9.498e-01  1.627e-02 -58.377 < 2e-16 ***
# Q14C                  -1.761e+00  2.493e-02 -70.641 < 2e-16 ***
# Q14DorF               -2.512e+00  7.553e-02 -33.264 < 2e-16 ***
# Q14other              -1.553e+00  7.314e-02 -21.230 < 2e-16 ***
# semesterspring:cap    -5.299e-04  2.412e-04 -2.197 0.028008 *
# course.levelLD:cap     3.779e-03  8.903e-04  4.245 2.19e-05 ***
# course.levelUD:cap     1.155e-03  9.154e-04  1.262 0.207008
# deptsociology:cap     -7.016e-04  1.264e-03 -0.555 0.578954
# depttechnology:cap     1.567e-04  5.846e-04  0.268 0.788726
# deptanthropology:cap  -6.652e-03  3.312e-03 -2.008 0.044639 *
# deptastronomy:cap     7.801e-02  2.922e-02  2.670 0.007594 **
# deptbiology:cap       -5.907e-04  5.789e-04 -1.020 0.307538
# deptnature:cap        3.479e-03  1.547e-03  2.249 0.024504 *
# deptbusiness:cap      -3.669e-04  7.968e-04 -0.460 0.645209
# deptchemistry:cap     -1.785e-03  6.267e-04 -2.849 0.004389 **
# deptcommunication:cap  2.705e-02  1.234e-02  2.193 0.028339 *
# depteconomics:cap     3.777e-03  1.666e-03  2.267 0.023409 *
# deptgeography:cap     -1.017e-02  1.211e-02 -0.840 0.400982
# depthistory:cap       -8.384e-03  1.555e-03 -5.393 6.93e-08 ***
# deptmeteorology:cap   4.599e-03  4.867e-03  0.945 0.344681
# deptedu:cap           -8.346e-03  9.298e-04 -8.976 < 2e-16 ***
# deptphysics:cap       -2.265e-03  7.882e-04 -2.874 0.004052 **
# deptpsychology:cap    -7.283e-05  6.651e-04 -0.110 0.912806
# depturban.planning:cap 1.926e-03  3.410e-02  0.056 0.954952
# ---
#   Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
#
# (Dispersion parameter for binomial family taken to be 1)
#
# Null deviance: 115734 on 84063 degrees of freedom
# Residual deviance: 105408 on 84014 degrees of freedom
# (2106 observations deleted due to missingness)
# AIC: 105508
#
# Number of Fisher Scoring iterations: 4

```

```

####model adequacy checking###
##half normal plot:
hnp(sotemodel3, resid.type = "standard", sim = 99, conf = 0.95, how.many.out = TRUE, print.out =
TRUE, paint.out = TRUE,cex=.1,lty=1:2,colour=c("skyblue","black"))
##cook's d
plot(cooks.distance(sotemodel3),type="h",ylab = "Cook's Distance")
indx <- which(cooks.distance(sotemodel3) > 0.008)
text(cooks.distance(sotemodel3)[indx] ~ indx, labels = indx,cex = 0.7, pos = 4, offset = .4)
points(indx, cooks.distance(sotemodel3)[indx], cex = .4, bg = cols[indx])
as.data.frame(sote[c(6911,62535),])

# semester grade grading course.level      dept college student.level enroll cap registered  Q13 Q14 Q15
# fall   other      CMC          GR meteorology   sci          gr      7  10      7 TRUE  A   5
# fall   other      CMC          GR meteorology   sci          gr      7  10      7 TRUE  A   5
n=length(cooks.distance(sotemodel3))
indx2 <- which(cooks.distance(sotemodel3) > 4/n)
##confusion matrix
confusion <- function(m, threshold = 0.5) {
  pred <- ifelse(m$fitted.values > threshold, 1, 0)
  confused <- matrix(NA, nrow = 2, ncol = 3)

  indx <- (m$y == 1)
  confused[1, 3] <- sum(indx)
  confused[2, 3] <- sum(!indx)

  confused[1, 1] <- sum(m$y[indx] == pred[indx])
  confused[1, 2] <- confused[1, 3] - confused[1, 1]
  confused[2, 2] <- sum(m$y[!indx] == pred[!indx])
  confused[2, 1] <- confused[2, 3] - confused[2, 2]

  rownames(confused) <- c("1", "0")
  colnames(confused) <- c("1", "0", "total")

  cat("Sensitivity =", confused[1, 1] / confused[1, 3], "\n")
  cat("Specificity =", confused[2, 2] / confused[2, 3], "\n")
  confused
}
confusion(sotemodel3,.2)
library(ROCit)
roc <- rocit(score = sotemodel3$fitted.values, class = sotemodel3$y)
plot(roc, YIndex = FALSE)
roc$AUC
###conclusion###
predict(sotemodel3,newdata =
data.frame(semester="fall",grading="GRD",course.level="UD",dept="edu",cap=10,Q14="A"),type="respo
nse",se.fit=T)
prop.test(sum(sote$Q13,na.rm=T),nrow(sote))
predict(sotemodel3,newdata =
data.frame(semester="fall",grading="GRD",course.level="UD",dept="math",cap=10,Q14="A"),type="resp
onse",se.fit=T)

```